Dig Packet Reader - Setup and User Guide

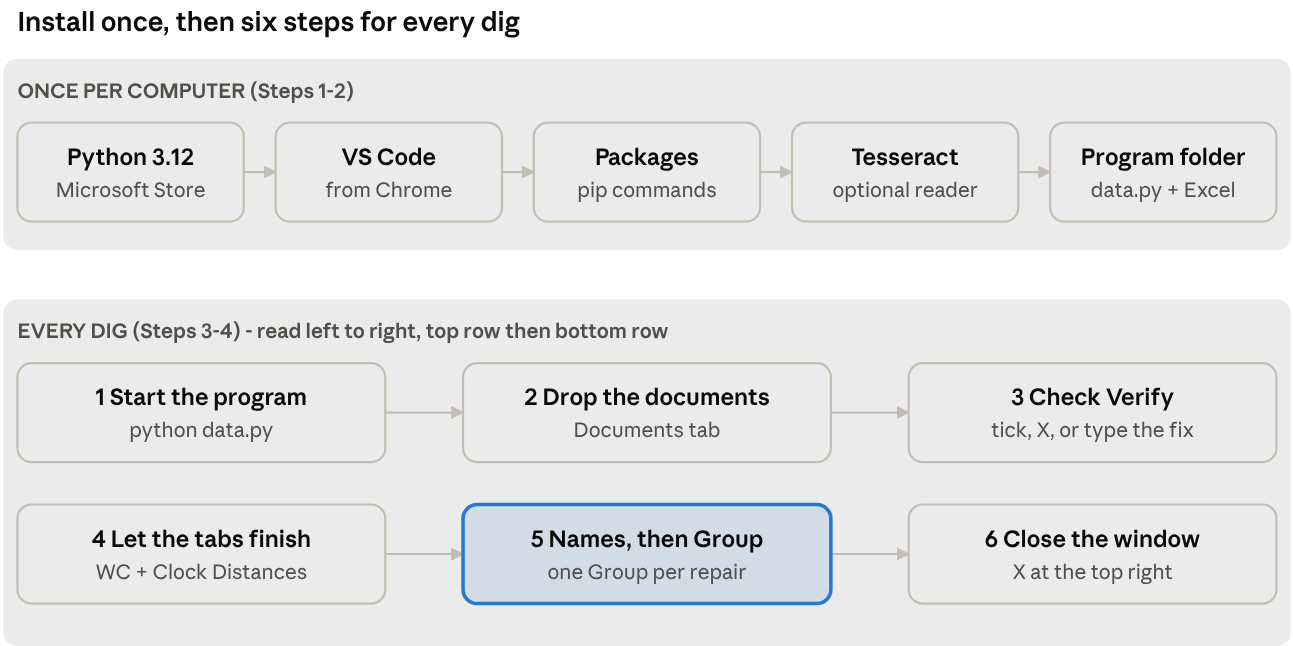
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How to install, start and use data.py (combined_6000_dig.py, version S466) from nothing, so anyone can run it.

This program was made by Maxim Dessiatov with Claude and ChatGPT.

If anything goes wrong: copy everything in the terminal (click in it, press **Ctrl+A** then **Ctrl+C**, or select the text and right-click) and paste it into a chat with **Claude**, together with what you were doing. That gives the best fix. Pasting the terminal after a normal run helps too: it shows Claude the timings and what was read, so it can improve the program or find problems you did not notice.

At a glance



The whole job at a glance - setup once, then 6 steps per dig

The top row is done once per computer (Steps 1-2); the six numbered steps are done for every dig (Steps 3-4). Step 5, Group, is the goal of each run.

What it does

You drop a dig's paperwork into one window. The program reads every document, lets you check what it read, works out wheel-count distances and repair lengths through the Excel composite calculator, and builds a ready-to-send Group folder with a filled-in calculator copy.

Documents it reads:

- **6000-10 Pipe Evaluation Information forms** (PDF)
- **Anomaly Investigation Dig Packets** (scanned PDF with red handwriting, map page, vendor dig sheet, ILI table screenshot)
- **10000-35 Pipe Wall Thickness Survey** workbooks (.xlsx), including corrosion / gouge grids
- **24-Hour Field Reports** (PDF)
- **OmegaWrap quote requests** (PDF)
- Any other PDF, picture (.png / .jpg) or email (.eml / .msg) - read as well as it can

What you get out: a checked table of every important value (OD, wall thickness, grade, depth, length, width, clock position, wheel count, coordinates and more), wheel-count and clock-position distances from the calculator, the repair the program picks, and a Group folder in Downloads per repair.

The program never does wheel-count math itself. Every distance, length and middle comes from the calculator file; the program only fills its input cells, reads its output cells and compares them.

Before you start

Everything used is free except Windows and Excel, which the work computer already has.

Need	Why	Notes
Windows 10 or 11 PC	The program uses Excel and Windows paths	Run as your normal user, no admin needed for most steps

Need	Why	Notes
NVIDIA graphics card (GPU) with current drivers	The handwriting / scan readers (EasyOCR, PaddleOCR) are set to run on the GPU	The original PC has an RTX A5000 Laptop GPU, 16 GB. Without an NVIDIA GPU, Dig Packets will not read
Microsoft Excel (desktop)	WC Distances, Clock Distances and repair lengths are worked out in the calculator	Must be the installed desktop Excel, not Excel online
Microsoft Edge	Opens the Google Maps links and place lookups	Already on Windows
Internet the first time	Downloads the packages and the OCR models (a few GB)	After the first run the models stay on the PC
The program file data.py	The whole program is this one file	Also called combined_6000_dig.py; current version S466
The calculator file	Any Excel file whose name starts with AE_Composite_Calculator	e.g. AE_Composite_Calculator_v2.06_072926 NO TimeWarp (2).xlsm
The documents	What gets read	Usually saved in Downloads

Step 1 - Install the software

Do these once per computer, in this order. Every command is typed in **PowerShell** (Start menu → type `PowerShell` → open **Windows PowerShell**). Paste a command, press Enter, wait until the prompt `PS C:\...>` comes back before the next one.

Fastest way if the original PC still works: on that PC run `python -m pip freeze > C:\Users\<you>\Downloads\requirements.txt`, copy that file to the new PC, and after steps 1.1-1.3 run `python -m pip install -r requirements.txt`. That installs the exact same versions. Then do the two checks at the end of 1.4 and skip to 1.6. If that install stops with an error on torch or paddlepaddle-gpu, install those two the 1.4 way and run the requirements command again.

1.1 Python 3.12

The original PC uses **Python 3.12 from the Microsoft Store**, so that is the way to match it:

1. Open the **Microsoft Store** app → search **Python 3.12** → the one published by the **Python Software Foundation** → **Get** / Install. No PATH box to tick; the Store sets it up.
2. Open a **new** PowerShell window and check: `python --version` should print `Python 3.12.x`.
3. If typing `python` opens the Store instead, turn that off: Windows Settings → Apps → Advanced app settings → **App execution aliases** → keep **python.exe** and **python3.exe** for Python 3.12 on, and make sure no other Python entry is on.

The python.org installer also works (tick **Add python.exe to PATH** on its first screen); just use one or the other, not both, so `python` and `pip` point at the same Python.

Store Python notes: packages install into your own user folder (no admin needed), and helper programs such as py-spy land in a Scripts folder that is not on PATH — the program looks there by itself, so nothing to change.

1.2 Visual Studio Code (to open and run the file)

1. Open **Google Chrome** (that is how it was done on the original PC), search **Visual Studio Code**, and open the official Microsoft site **code.visualstudio.com** (not an ad or another download site).
2. Click **Download for Windows**. The installer lands in Downloads (named like `VSCodeUserSetup-x64-....exe`).
3. Double-click it, accept the agreement, and keep the default options (leave "Add to PATH" ticked) → **Install** → **Finish**.
4. Open VS Code → Extensions (the four-squares icon on the left bar, or **Ctrl+Shift+X**) → search **Python** → install the one by **Microsoft**.
5. Tell VS Code which Python to use: press **Ctrl+Shift+P** → type **Python: Select Interpreter** → pick the **Python 3.12** entry. Do this once; VS Code remembers it.

1.3 Update pip

```
python -m pip install --upgrade pip
```

1.4 GPU readers (the big downloads)

These two must match the computer's NVIDIA CUDA version, so take the command from each site's install picker instead of typing a fixed one:

First find the CUDA version the graphics card supports. In PowerShell run:

```
nvidia-smi
```

It prints a box with the card name (e.g. NVIDIA RTX A5000 Laptop GPU) and, top right, **CUDA Version: 12.x**. Pick that number, or the closest lower one, in the two pickers below. If `nvidia-smi` is not recognized, install the NVIDIA driver first from nvidia.com/drivers, restart the PC, and try again.

1. **PyTorch** (used by EasyOCR and the language model): go to pytorch.org/get-started/locally, choose Stable · Windows · Pip · Python · the newest CUDA listed, copy the command it shows (it starts `pip3 install torch torchvision ...`) and run it.
2. **PaddlePaddle GPU** (used by PaddleOCR): go to paddlepaddle.org.cn/en/install/quick, choose Windows · pip · GPU · the CUDA version, and run the command it shows (it installs `paddlepaddle-gpu`).
3. Check both see the GPU:

```
python -c "import torch; print(torch.cuda.is_available())"
python -c "import paddle; print(paddle.device.is_compiled_with_cuda())"
```

Both must print `True`. If not, update the NVIDIA driver from nvidia.com/drivers and reinstall that package.

1.5 The other Python packages

One command, all free:

```
python -m pip install pymupdf numpy opencv-python pillow wordfreq easyocr
paddleocr transformers pywin32 tkinterdnd2 openpyxl pgeocode us playwright
extract-msg py-spy
```

Package	What it is for	If missing
pymupdf, numpy, opencv-python, pillow, wordfreq	Reading PDFs, pictures, word lists	The program will not start

Package	What it is for	If missing
easyocr, paddleocr	Reading handwriting, scans and the ILI table pictures	Dig Packets cannot be read
transformers	Language model that picks between close handwriting reads	Downloads its model the first time it is needed
pywin32	Talks to Excel (calculator, WC / Clock Distances)	Terminal says `needs: pip install pywin32`
tkinterdnd2	Drag-and-drop into the window	Window says drag-and-drop is not available
openpyxl	Reads 10000-35 .xlsx forms	.xlsx files are not read
pgeocode, us	Checks the town / state typed for the Group folder name	The town check is skipped
playwright	Keeps one hidden Edge open for faster Google Maps lookups	Each lookup opens Edge on its own (slower)
extract-msg	Reads Outlook .msg emails	.msg files are not read
py-spy	Records a speed profile (profile.svg) of each run	Terminal says the run is not recorded - harmless

1.6 Tesseract (second reader for printed table cells)

1. Download the Windows installer from github.com/UB-Mannheim/tesseract/wiki.
2. Install with the default folder `C:\Program Files\Tesseract-OCR`. The program finds it there by itself.
3. Without it the program still runs; table cells are then read by PaddleOCR / EasyOCR only.

1.7 Excel macros warning

Nothing to set. The program opens its own hidden copy of the calculator with macros off and closes it again; your open Excel windows are not touched.

Step 2 - Set up the folders

The program and the calculator live in one folder; the documents can be anywhere (usually Downloads).

Getting the two files. Both come from the original PC's program folder, `C:\Users\MaximDessiatov\python`: copy **data.py** and the **AE_Composite_Calculator…** Excel file onto a USB stick, OneDrive or a shared drive, or ask Maxim Dessiatov to send them. A newer data.py may also come as a file named `combined_6000_dig.py` downloaded from Maxim Dessiatov's Claude chat. Always use the newest one (the most recent date in File Explorer's Date modified column); if unsure, check with Maxim Dessiatov.

1. Make the program folder: `C:\Users\<your user name>\python` (on the original PC: `C:\Users\MaximDessiatov\python`). In File Explorer: open `C:\Users\<you>` → right-click → New → Folder → name it `python`.
2. Put **data.py** in it. If you received it as `combined_6000_dig.py`, rename it to `data.py` (either name runs the same).
3. Put the **calculator** in the same folder. Any Excel file whose name starts with `AE_Composite_Calculator` works (`.xlsm`, `.xlsx`, `.xlsb`, `.xls`). If there are several, the most recently saved one is used. Do not rename it to something else and do not keep it open in Excel while the program runs.
4. Documents: save the 6000 forms, Dig Packets, 10000-35 workbooks, 24hr reports and quotes anywhere, e.g. `C:\Users\<you>\Downloads`. You will drag them into the window.

When it is done the folder looks like this:

```
C:\Users\<you>\python\
    data.py
    AE_Composite_Calculator_v2.06_072926 NO TimeWarp (2).xlsm
```

On a different user account: the calculator is found next to data.py, and Group folders go to that user's own Downloads, so nothing needs editing. Only the debug-picture folder near the top of data.py (`DOWNLOAD_FOLDER`, line ~381) still names `C:\Users\MaximDessiatov\Downloads`; it is only used for optional debug pictures (see Settings).

Step 3 - Start the program

Either way works; keep the terminal (black text window) open the whole time, because it shows progress and any problems.

From PowerShell

```
cd C:\Users\<you>\python
python data.py
```

From VS Code

1. File → **Open Folder…** → pick `C:\Users\<you>\python` → Select Folder (say Yes, trust the authors, if asked).
2. Click `data.py` in the list on the left.
3. Terminal → **New Terminal** (bottom panel opens) → type `python data.py` → Enter. (The ▷ Run button at the top right does the same.)

The VS Code terminal is the same PowerShell, so every command in this guide can also be typed there. If the terminal says `python` is not found, close and reopen VS Code once after installing Python.

What happens next

- A window called the **Documents** window opens with a big "Drag files here" box.
- The terminal prints lines such as `PROFILE: py-spy is recording this run...` and, once documents are read, `CALCULATOR: using C:\...\AE_Composite_Calculator....xlsm`.
- **First run only:** the OCR models download when the first Dig Packet is read (a few minutes and a few GB). Later runs start in seconds; the models load in the background the moment a Dig Packet is dropped in.

Step 4 - Use the window

Work top to bottom: drop the documents, check Verify, look at the distance tabs, fill in the names, click Group. A tab that is still working is greyed out with **(loading)** in its name and cannot be opened until it is ready.

1. **Drop the documents.** Open the **Documents** tab and drag all of one dig's files from File Explorer onto the "Drag files here" box (6000 forms, Dig Packets, 10000-35 .xlsx, 24hr report, quote, emails). You can drop more later. Each file gets a row with its progress

(`step 3/9 …`) and turns green when read; the **X** on a row removes that file.

2. **Check Verify.** The **Verify** tab shows the merged PDF of everything dropped and jumps to each value read from handwriting or a picture, boxed, with a small panel under it:
   - ✓ = read correctly, go to the next one.
   - **X** = bad read, ignore it.
   - Type in the box = the correct value.
   - Values that are PDF text are exact and are not asked.
3. **Pick between documents that disagree.** When two documents give different values (an OD, a wall thickness, a length…), Verify lists them; click ✓ on the right one (it turns green). The **WC Distances** tab waits with "(waiting for Verify picks)" until these are done.
4. **Read the end table.** Under Verify comes one column per anomaly (headed by its wheel count) with every priority value and "Everything else found". Yellow cells disagree; the "Final" column holds your pick.
5. **WC Distances / Main Wheel Counts.** Distances between neighbouring wheel counts, repair lengths and combined repairs, all worked out by the calculator in Excel, plus **The repair it picks** (fabric: E-Glass only for pure corrosion / metal loss, Carbon otherwise; Carbon adds the +12 in isolation wrap) and the wrap checks.
6. **Clock Distances.** Distance around the pipe between anomalies with a clock position and width, and to the long seam, from the calculator's Clock Calculator sheet. Text in the warning cells means two of them overlap.
7. **Warnings.** Anything odd the program noticed (a wheel count missing from a table, overlapping wraps…), each with the snip it came from.
8. **Document names (on the Verify tab, above the end table).** One row per repair:
   - Check or retype the name in the box (it starts as the bold main wheel count).
   - Tick **Right?** → the rest of the row appears.
   - Type the **overall company** (optional) and **company name**.
   - Click **Google Maps**, copy the place name Google shows (e.g. `FWXF+3QJ McFaddin, Texas`) into the place box; the plus code is removed by itself. Type the **state** abbreviation.
   - The folder name and document name fill in as you type.
9. **Click Group.** The button stays grey and says "Group (waiting for WC Distances, Clock Distances)" until both tabs are finished; then it turns green. Click it once per repair.

Other tabs: each document gets its own tab(s) with everything read from it and its snips; **Pages** shows each page with what was read marked; **Extra** lets you open or close extra per-document tabs (✓ shows, X hides).

The Group folder

Each Group click makes one new folder in **Downloads**, named with the folder name from that row. An existing folder is never replaced: a second one gets `(2)`, then `(3)`. When it worked the button changes and the terminal prints `GROUP: <repair> -> <folder>`.

File in the folder	What it is
`Composite_Calc_<…>.xlsm`	Copy of the calculator, renamed to the document name, with the cells below filled in
`Composite_Supporting Docs_<…>.pdf`	Every PDF of the repair's anomalies merged into one
`All Documents.pdf`	Everything you dropped in, merged (the same PDF Verify shows)
The original files	A copy of each document the repair's values came from
`profile.svg`	Speed profile of the run, copied in when the window is closed (only if py-spy is installed)

Cells written in the calculator copy:

Sheet	Cell	Value
Main	C3	Diameter, wheel count (or station), Line ID, project number - e.g. `20", WC 100553.07, SHC 20-1, PN 17748`
Main	C9	City, State from the place / state boxes
Main	H4	The date Group was clicked, mm/dd/yyyy
Main	H9	Latitude, longitude
Main	H10	Defect name

Sheet	Cell	Value
Main	C11	Only for a repair covering several anomalies: "One wrap covers WC … L …, W …, D …, …"
Main	C15	Corrosion (only corrosion / metal loss) or Mechanical (anything else)
Corrosion and Mechanical	E4, E5	Outside diameter, wall thickness (inches)
Corrosion and Mechanical	E7	Pipe grade (only when a document says "Pipe Grade")
Corrosion and Mechanical	E10	Class location as 72 / 60 / 50 / 40 (Class 1 assumed when not given)
Corrosion and Mechanical	E13, E15, E16	Largest depth, total defect length, largest width (inches)
Corrosion E19 / Mechanical E18	fabric	Carbon or E-Glass

Open the calculator copy in Excel to finish the repair design as usual.

Closing the program

Close the window with its **X** (top right) when you are done; the terminal then finishes on its own and the prompt comes back.

- It closes its hidden Excel copies of the calculator and deletes them, and deletes its temporary merged PDFs.
- It writes `profile.svg` to Downloads (and into any Group folder made that run) and prints a summary at the end: `FINAL TIME`, `VERIFY WAIT`, `OCR USE`, `GPU MEMORY`. These are information only.
- For the next dig, start it again (Step 3). Don't leave it open overnight: the speed recorder falls behind while the PC sleeps and fills the terminal with "behind in sampling" lines (harmless).
- If the window will not close, click in the terminal and press **Ctrl+C**.

The only things left on the PC are the Group folders, `Downloads\profile.svg` (replaced each run) and the downloaded OCR models.

When something goes wrong

Look at the terminal first; nearly every problem prints a line saying what is missing.

Best fix for any error: copy the whole terminal and paste it into Claude with a sentence on what you did (which documents, which button). The table below covers the common cases; anything not in it, paste it to Claude.

What you see	What it means	What to do
`'python' is not recognized` or the Microsoft Store opens	Python not installed or not on PATH	Step 1.1 (App execution aliases for the Store version)
`ModuleNotFoundError: No module named 'fitz'` (or cv2, numpy, wordfreq…)	A package is missing	Run the Step 1.5 command again
Window says "Drag-and-drop is not available"	tkinterdnd2 missing	`python -m pip install tkinterdnd2`, start again
Dig Packet stuck or `CUDA` / `gpu` errors	GPU readers not set up	Step 1.4 checks must both print True; update the NVIDIA driver
`CALCULATOR: no 'AE_Composite_Calculator...' Excel file in …`	Calculator not next to data.py	Put it in the program folder (Step 2)
WC Distances says `needs: pip install pywin32`	pywin32 missing	`python -m pip install pywin32`
WC Distances says Excel could not work them out	Excel blocked or busy	Close any Excel dialog, make sure the calculator is not open, start again
WC Distances shows "(waiting for Verify picks)"	Two documents disagree on a value	Finish the picks on Verify
Group button grey	WC Distances / Clock Distances still working	Wait until both tabs open
"Group - name missing"	Folder or document name empty	Tick Right? and fill the boxes

What you see	What it means	What to do
"Group - failed"	Folder could not be made	Read the `GROUP:` line in the terminal (often a file open in another program)
`TESSERACT: not installed`	Optional second table reader missing	Step 1.6, or ignore
`PROFILE: py-spy not found`	Optional speed recorder missing	`python -m pip install py-spy`, or ignore
`py-spy> … behind in sampling` many times	The PC slept with the program open	Harmless; close and restart the program
Excel opens the Group calculator copy in Protected View or says macros are blocked	Windows marked the calculator as downloaded from the internet	Click Enable Editing. If macros stay blocked: close it, right-click the original calculator in the python folder → Properties → tick Unblock → OK; new Group copies then open normally

A value read wrong: correct it on **Verify** (type the right value). It only changes this run; the program file is not changed.

Settings in data.py

Normally nothing needs changing. If something must, open data.py in VS Code, press **Ctrl+G**, type the line number, change only the value, save with **Ctrl+S**, and start the program again. Line numbers are for version S466.

Line	Setting	Now	Change it when
381	`DOWNLOAD_FOLDER`	`C:\Users\MaximDessiatov\Downloads`	On another user account (only used for optional debug pictures)
2776	`OCR_PROCESS_ON`	`True`	Leave True; the OCR runs in its own process so the window stays responsive
22583	`TESSERACT_PATHS`	`C:\Program Files\Tesseract-OCR\tesseract.exe` and two others	Tesseract was installed somewhere else
29694	`PIPE_GRADES`	24,000, A25, A, B, X42 … X70	The calculator's Pipe Grade dropdown changes
29702	`CLASS_LOCATION_VALUES`	Class 1 → 72, 2 → 60, 3 → 50, 4 → 40	The calculator's Class Location dropdown changes

Line	Setting	Now	Change it when
29831	`GROUP_DATE_CELL`	`H4`	The date moves to another cell on Main
30380	`MAPS_BROWSERS`	Edge in Program Files	Edge is installed somewhere else
30562	`CALCULATOR_PREFIX`	`AE_Composite_Calculator`	The calculator file gets a new name pattern
30564	`CALCULATOR_FOLDERS`	the data.py folder, then `C:\Users\MaximDessiatov\python`	The calculator should be kept in another folder
35506	`PROFILE_RUN`	`True`	Set `False` to stop recording profile.svg

Calculator cell addresses (which cell gets the wheel counts, lengths, fabric…) are also named near line 29600-29840; change them only if the calculator's layout changes, and keep a copy of the working data.py first.